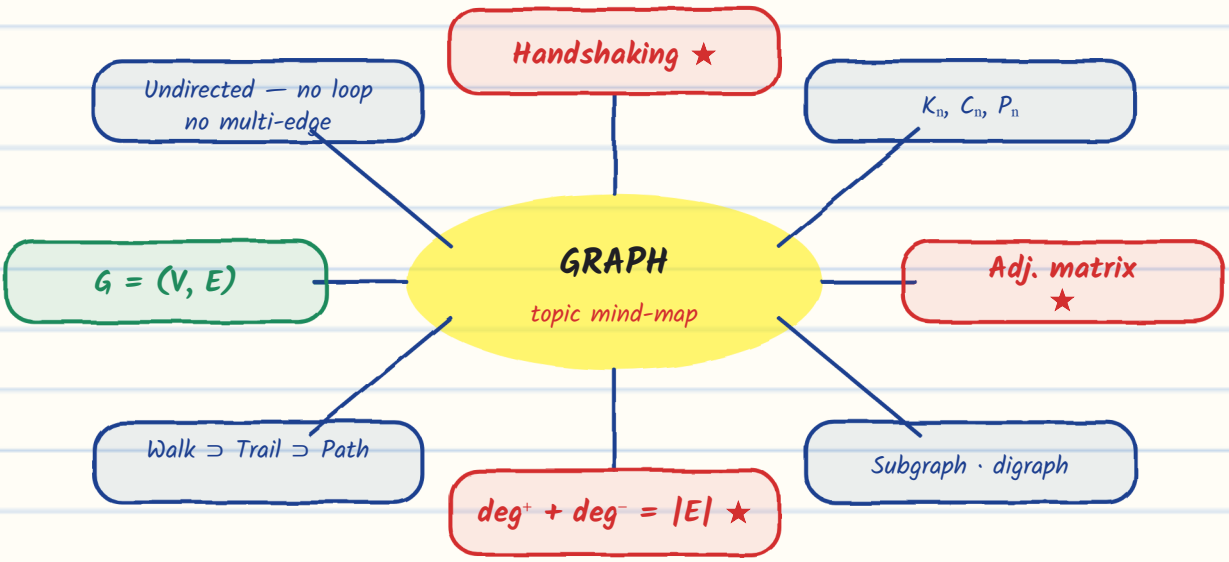
Date: 13.10.2026
topper's notes

Graph Fundamentals

G = (V,E) · simple vs directed · degree · walk/path/cycle · K_n/C_n/$K_{m,n}$: exam-ready in 10 sheets



↑ the whole topic on one page: red = high-yield, green = the one definition every question builds from

➜ Syllabus coverage

The syllabus line is "Graph theory: basic definitions — simple graph, directed graph, subgraph, degree, walk, path, cycle, special graphs (K_n, C_n, $K_{m,n}$), representations (adjacency matrix, adjacency list)". This lesson covers the foundation that Trees, Euler & Hamiltonian, and Planar Graphs lessons build on.

High-yield list

① G=(V,E) def
② no loop, no multi-edge
③ $deg^+ + deg^- = |E|$
④ Handshaking: $\Sigma deg = 2|E|$
⑤ Walk ⊇ Trail ⊇ Path ⊇ Cycle
⑥ K_n has $n(n-1)/2$ edges
⑦ Adj. matrix $O(V^2)$
⑧ Cycle C_n has n edges

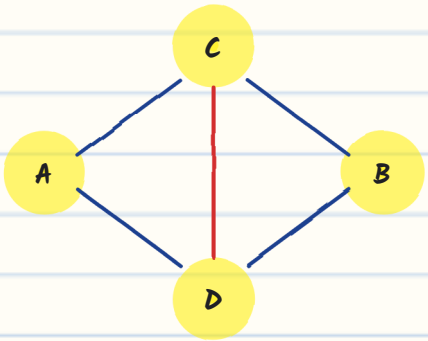
Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

➜ Contents

✓ What is a graph? Simple vs directed ... p. 2
✓ Subgraphs & vertex degrees ... p. 3
✓ Walks, trails, paths, cycles ... p. 4
✓ Special graphs K_n, C_n, P_n ... p. 5
✓ Bipartite & regular graphs ... p. 6
✓ Graph representation ... p. 7
✓ Degree sequences & Havel-Hakimi ... p. 8
✓ Night-before p. 9 · Practice p. 10

1 What Is a Graph? definition ★

Definition A graph G is an ordered pair $G = (V, E)$, where V is a non-empty set of vertices (nodes) and E is a set of edges (links). In an **undirected** graph each edge is an unordered pair $\{u, v\}$; in a **directed graph** (digraph) each edge is an ordered pair (u, v), written $u \to v$.



left: undirected $\{A,B\}$ … right: directed $A \to B$

undirected: edge $\{A,B\}$ same as $\{B,A\}$ · directed: edge $(A,B) \neq (B,A)$

Simple graph An undirected graph with **no loops** (an edge from a vertex to itself) and **no multiple edges** between the same pair of vertices. Every pair of distinct vertices is joined by at most one edge.

Exam trap (i) A **loop** at vertex v is an edge $\{v, v\}$ — it breaks simplicity. (ii) **Multiple edges** between u and v also break simplicity. A graph with either is a multigraph, not a simple graph. (iii) "Simple" does **not** mean "small" — K_{100} is still a simple graph.

Term	Undirected	Directed (digraph)
Edge	$\{u, v\} = \{v, u\}$	$(u, v) \neq (v, u)$ in general
Loop	$\{v, v\}$ — forbidden in simple graphs	(v, v) — self-loop
Multiple edge	Two edges $\{u,v\}$ and $\{u,v\}$ — forbidden	Two edges (u,v) and (u,v) — forbidden in simple digraph

Memory trick "**Simple = clean.**" No self-loops, no double edges — just pure connections between distinct vertices. The moment you allow loops or parallel edges, you leave the world of simple graphs and enter multigraphs.

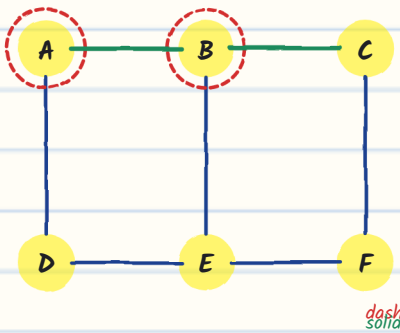
Worked example ✍ is this simple?

$V=\{A,B,C\}$, $E=\{\{A,B\},\{B,C\},\{C,A\},\{A,A\}\}$. The edge $\{A,A\}$ is a **loop** — so **not** simple. Remove it and the remaining graph **is** simple (it is the triangle C_3).

2 Subgraphs & Vertex Degrees

foundational ★

Definition A **subgraph** H of $G = (V, E)$ is a graph $H = (V', E')$ such that $V' \subseteq V$, $E' \subseteq E$, and every edge in E' has both endpoints in V'. A **spanning subgraph** is a subgraph with $V' = V$ (it keeps all vertices but may drop some edges).



H keeps A–B and C–D (renamed) edges; D and F are isolated in H

Concept	Undirected	Directed
Degree $\deg(v)$	# of edges incident to v	—
In-degree $\deg^-(v)$	—	# of edges entering v
Out-degree $\deg^+(v)$	—	# of edges leaving v

Handshaking Lemma — undirected Σ $\sum_{v} \deg(v) = 2|E|$. Every edge touches exactly two ends, so it is counted twice in the degree sum.

Corollary: the number of odd-degree vertices is always even.

Handshaking Lemma — directed Σ $\sum_{v} \deg^+(v) = \sum_{v} \deg^-(v) = |E|$. Every directed edge leaves one vertex and enters another, so total out-degrees = total in-degrees = number of edges.

Memory trick "**Every edge has two hands to shake.**" Count every handshake from each vertex's perspective — each edge is shaken at both ends, so it contributes 2 to the total. Same idea for digraphs: each edge shakes once from the tail and once at the head.

Exam trap (i) Handshaking holds for **multigraphs** too — the "simple" condition is not needed for the lemma. (ii) A loop in an undirected graph contributes **2** to the degree of its vertex (it touches the vertex at two ends). (iii) $\sum \deg^+ = \sum \deg^- = |E|$ for digraphs; do not multiply by 2 here.

Worked example ✍ applying Handshaking A simple undirected graph has 12 edges. By the lemma, $\sum \deg(v) = 2 \times 12 = \mathbf{24}$. If the graph has 6 vertices, the average degree is 24/6 = 4. Can every vertex have degree 4? Yes — that would be a 4-regular graph on 6 vertices (e.g. the complement of a perfect matching).

3 Walks, Trails, Paths & Cycles

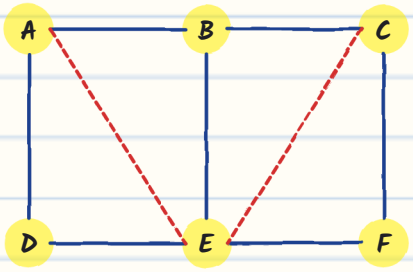
terminology ★

Definition Given a graph G, a **walk** of length k is a sequence $v_0, e_1, v_1, e_2, \ldots, e_k, v_k$ where each e_i joins v_{i-1} and v_i. Vertices and edges **may repeat**. A **trail** is a walk with **no repeated edge** (vertices may repeat). A **path** (or simple path) is a walk with **no repeated vertex** — hence no repeated edge automatically. A **cycle** is a path that starts and ends at the same vertex; for a simple graph the length must be ≥ 3.

Type	Vertices repeat?	Edges repeat?	Closed?
Walk	yes	yes	maybe
Trail	yes	no	maybe
Path	no	no	no (unless trivial)
Cycle	only start=end	no	yes

Inclusion diagram

Walk — least restrictive
↓ no repeated edge
Trail
↓ no repeated vertex
Path
↓ closed
Cycle — most restrictive



$A \to B \to E \to D \to B \to C$ is a walk but not a path (B repeats)

Every path is a trail is a walk. Not every walk is a trail; not every trail is a path.

$v_0 e_1 v_1 e_2 \ldots e_k v_k$ — repeat freedom decreases as you move down the hierarchy

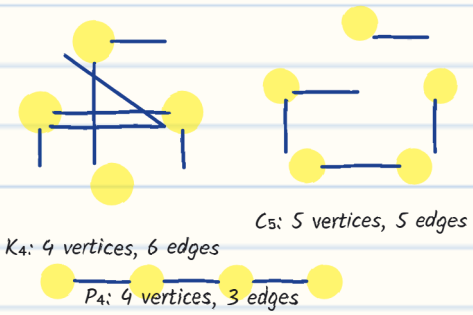
Exam trap (i) "Path" in everyday language means any route; in graph theory it means **no repeated vertex**. (ii) A cycle must have **length ≥ 3** in a simple graph — two vertices with two edges between them (a multigraph) would be a cycle of length 2, but that is impossible in a simple graph. (iii) A single vertex with a self-loop is **not** a cycle in a simple graph (loops are forbidden).

Memory trick "**No vertex repeated = path; no edge repeated = trail; both allowed = walk; closed path = cycle.**" Remember: path is the strictest common type; cycle is a path that returns home.

4 Special Graphs I: Complete, Cycle, Path

formulas ★

Graph	Structure	|E|	Degree of each vertex
K_n	every pair joined	$n(n-1)/2$	$n-1$
C_n	n vertices in one cycle	n	2
P_n	n vertices in a line	$n-1$	end = 1, internal = 2



K_4 is 3-regular; C_5 is 2-regular; P_4 has two endpoints of degree 1

Conditions K_n requires $n \geq 1$ (K_1 has 0 edges). C_n requires $n \geq 3$ (a 2-cycle needs parallel edges, forbidden in simple graphs). P_n requires $n \geq 1$ (P_1 is a single vertex).

k-Regular graph A graph where **every** vertex has the same degree k. K_n is $(n-1)$-regular. C_n is 2-regular. $K_{n,n}$ is n-regular. For a k-regular graph on n vertices, $|E| = nk/2$, so **nk must be even**.

k-Regular graph A graph where **every** vertex has the same degree k. K_n is $(n-1)$-regular. C_n is 2-regular. $K_{n,n}$ is n-regular. For a k-regular graph on n vertices, $|E| = nk/2$, so **nk must be even**.

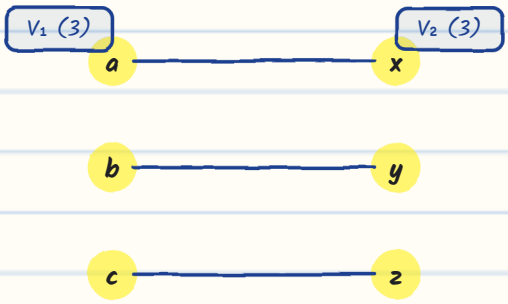
Exam trap (i) K_1 has 0 edges, K_2 has 1 edge. K_3 is the triangle C_3. (ii) C_n exists only for $n \geq 3$ — C_1 and C_2 are not simple graphs. (iii) P_1 is a single vertex (0 edges); P_2 is a single edge. (iv) **K_n is $(n-1)$-regular** — every vertex has the same degree.

Memory trick **"K gets all connections, C makes a ring, P draws a line."** K_n: everyone shakes hands with everyone. C_n: stand in a circle holding hands. P_n: stand in a queue.

5 Bipartite & Regular Graphs $K_{m,n}$ ★

Definition A **bipartite graph** has its vertices partitioned into two sets V_1 and V_2 such that every edge joins a vertex in V_1 to one in V_2 — no edge joins two vertices within the same set. A **complete bipartite graph** $K_{m,n}$ joins **every** vertex of V_1 to **every** vertex of V_2.

Property	$K_{m,n}$
$\|V\|$	$m + n$
$\|E\|$	$m \times n$
Degree of $v \in V_1$	n
Degree of $v \in V_2$	m
Regular?	Only when $m = n$ (then n-regular)



$K_{3,3}$: 9 edges, 6 vertices

every vertex on the left connects to every vertex on the right

k-Regular graph A graph where **every** vertex has the same degree k. K_n is $(n-1)$-regular. C_n is 2-regular. $K_{n,n}$ is n-regular.

For a k-regular graph on n vertices, $|E| = nk/2$, so **nk must be even**.

König's theorem A graph is **bipartite if and only if it has no odd cycle**.

Equivalently: G is bipartite $\iff$ every cycle in G has even length.

Odd cycle present $\Rightarrow$ not bipartite; all cycles even $\Rightarrow$ bipartite.

Exam trap (i) C_4 (square) **is** bipartite — vertices alternate between the two sets. C_5 (pentagon) is **not** bipartite — going around forces one vertex into both sets. (ii) $K_{2,3}$ is bipartite but **not** regular (degrees are 3 and 2). (iii) K_n is bipartite **only** for $n \leq 2$ — K_3 already contains a 3-cycle.

Memory trick "**Even cycle = bipartite; odd cycle = not.**" Check any cycle: if all are even-length, the graph is bipartite. One odd cycle destroys bipartiteness entirely.

6 Graph Representation

practical ★

Definition A graph must be stored in some data structure. Three standard representations trade off space, lookup speed, and ease of traversal. The choice depends on whether the graph is *dense* ($|E| \approx |V|^2$) or **sparse** ($|E| \ll |V|^2$).

Representation	Space	Edge lookup	Iterate neighbours	Best for
Adjacency matrix	$O(V^2)$	$O(1)$	$O(V)$	dense graphs
Adjacency list	$O(V + E)$	$O(\deg(v))$	$O(\deg(v))$	sparse graphs
Edge list	$O(E)$	$O(E)$	$O(E)$	algorithm input

Worked example ✍ a 4-vertex graph

G: $V=\{1,2,3,4\}$, $E=\{\{1,2\},\{1,3\},\{2,3\},\{2,4\}\}$.

Adjacency matrix (rows=cols=1,2,3,4):

```
0 1 1 0
1 0 1 1
1 1 0 0
0 1 0 0
```

Adjacency list:

```
1 → [2, 3]
2 → [1, 3, 4]
3 → [1, 2]
4 → [2]
```

Edge list:

```
(1,2) (1,3) (2,3) (2,4)
```

Exam trap (i) For a **directed** graph, the adjacency matrix need **not** be symmetric: $M[i][j]=1$ means $i \to j$, not $j \to i$. (ii) Adjacency matrix uses V^2 space **even for sparse graphs** — avoid for large sparse graphs. (iii) Self-loops: matrix diagonal entry = 1 (undirected) or 1 per loop direction (directed). (iv) Multiple edges: adjacency matrix cannot represent them — use adjacency list with counts.

Memory trick "**Matrix is square, list is lean.**" $O(V^2)$ space is the matrix's liability; $O(V+E)$ is the list's strength. For dense graphs the difference shrinks; for sparse graphs the list wins hands down.

7 Degree Sequences & Handshaking Applications

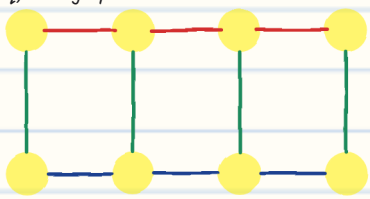
Havel-Hakimi ★

Definition The degree sequence of a graph is the list of vertex degrees written in non-increasing order, e.g. (3, 3, 2, 2, 1, 1). A sequence that can be realised as the degree sequence of **some** simple graph is called graphical.

Havel-Hakimi algorithm To test whether a sequence $d = (d_1, d_2, \ldots, d_n)$ is graphical:

1. Sort in non-increasing order.
2. Remove d_1 and subtract 1 from the next d_1 terms.
3. Re-sort and repeat.
4. If you ever get a negative → **not graphical**. If you reach all zeros → **graphical**.

same seq, diff graph



not isomorphic!

(3,3,1,1) → (2,0,0) → negative? no, but 2>remaining → fails

same degree sequence does not imply isomorphic graphs

Worked example ✍ Havel-Hakimi on (3, 3, 1, 1)

1. Remove first 3, subtract 1 from next 3: (3,1,1) → (2, 0, 0).
2. Remove first 2, subtract 1 from next 2: (0,0) → (-1, -1). **Negative → not graphical.**

In a simple graph on 4 vertices, a vertex of degree 3 is adjacent to **all** others. Two such vertices force every other vertex to have degree ≥ 2, contradicting the 1s.

Corollaries of Handshaking (i) $\Sigma \deg(v) = 2|E|$, so the sum is always **even**. (ii) The number of odd-degree vertices is **always even**. (iii) A regular graph of degree k on n vertices has $|E| = nk/2$, so **nk must be even**.

Exam trap (i) **Same degree sequence ≠ isomorphic.** Two graphs can share the same degree sequence yet have different structures. (ii) Havel-Hakimi tests for **simple** graphs only — it does not apply to multigraphs. (iii) A sequence with an odd sum can **never** be graphical (violates Handshaking).

IT link Degree-sequence analysis is used in network science to characterise real-world networks (social, biological, infrastructural) and compare them against random-graph models.

Night-Before-the-Exam Sheet

Fact sheet

- $G = (V, E)$: vertices & edges
- Simple: no loop, no multi-edge
- Digraph: edges are ordered pairs
- Subgraph: $V' \subseteq V$, $E' \subseteq E$, endpoints in V'
- $\deg(v)$: # incident edges
- $\deg^+(v)$: out-degree · $\deg^-(v)$: in-degree
- Handshaking: $\Sigma \deg(v) = 2|E|$
- Digraph: $\Sigma \deg^+ = \Sigma \deg^- = |E|$
- Walk ⊃ Trail ⊃ Path ⊃ Cycle
- K_n: $|E| = n(n-1)/2$, deg $= n-1$
- C_n: $|E| = n$, 2-regular
- P_n: $|E| = n-1$, ends deg 1
- $K_{m,n}$: $|E| = m \times n$
- Bipartite ⟺ no odd cycle
- Adj. matrix: $O(V^2)$ space, $O(1)$ lookup
- Adj. list: $O(V+E)$ space
- Havel-Hakimi: subtract & re-sort
- Odd-degree vertices: always even count

→ Top 12 one-liners for MCQs

1. $G = (V, E)$: V vertices, E edges
2. Simple = no loop, no multi-edge
3. Handshaking: $\Sigma \deg(v) = 2|E|$
4. Digraph: $\Sigma \deg^+ = \Sigma \deg^- = |E|$
5. Walk ⊃ Trail ⊃ Path ⊃ Cycle
6. K_n has $n(n-1)/2$ edges
7. C_n is 2-regular with n edges
8. $K_{m,n}$ has $m \times n$ edges
9. Bipartite ⟺ no odd cycle
10. Adj. matrix $O(V^2)$; list $O(V+E)$
11. Odd-degree vertices: even count
12. Same degree seq ≠ isomorphic

Top traps "path" in graph theory = no repeated vertex (not just any route) · cycle needs length ≥ 3 in simple graphs · Handshaking holds for multigraphs too · K_n is $(n-1)$-regular, not n-regular · adjacency matrix not symmetric for digraphs · same degree sequence ≠ isomorphic · C_n requires $n \geq 3$ · K_n bipartite only for $n \leq 2$

All tricks together "Simple = clean" · "Every edge has two hands to shake" · "K gets all connections, C makes a ring, P draws a line" · "Even cycle = bipartite; odd cycle = not" · "Matrix is square, list is lean" · "No vertex repeated = path; no edge repeated = trail; both allowed = walk; closed path = cycle".

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. Maximum edges in a simple graph with 10 vertices:
(a) 90 (b) 45
(c) 100 (d) 50

2. An undirected graph has 12 edges. Sum of all vertex degrees:
(a) 12 (b) 6
(c) 24 (d) 144

3. In any undirected graph, the number of odd-degree vertices is:
(a) always even (b) always odd
(c) always zero (d) equal to $|E|$

4. In a digraph, the sum of all in-degrees equals:
(a) $2|E|$ (b) $|V|$
(c) $|E|$ (d) $|E|^2$

5. A walk with no repeated edge but vertices may repeat is called a:
(a) trail (b) path
(c) cycle (d) bridge

6. Which sequence CANNOT be a simple graph degree sequence?
(a) (3,3,3,3) (b) (2,2,2,2)
(c) (3,3,1,1) (d) (2,2,1,1)

7. A 4-regular simple graph with 9 vertices has how many edges?
(a) 18 (b) 36
(c) 13 (d) 9

8. How many edges does K(4,5) have?
(a) 9 (b) 20
(c) 36 (d) 10

9. A graph is bipartite iff it has no:
(a) even cycle (b) odd cycle
(c) vertex of degree 1 (d) bridge

10. For a simple graph with 8 vertices and 10 edges, the complement has how many edges?
(a) 10 (b) 28
(c) 38 (d) 18

Answer key → 1-b · 2-c · 3-a · 4-c · 5-a · 6-c · 7-a · 8-b · 9-b · 10-d

Written question State and prove the Handshaking Lemma. Hence show that every undirected graph has an even number of vertices of odd degree. Illustrate with an example. 10

1. **Statement** — For any undirected graph $G = (V, E)$, $\Sigma_{v \in V} \deg(v) = 2|E|$. 2
2. **Proof** — Each edge $\{u, v\}$ has two ends. When summing degrees, that edge is counted once at u and once at v (a loop counts twice at its single vertex). So each edge contributes exactly 2 to the sum. Hence $\Sigma \deg(v) = 2|E|$. 4
3. **Corollary** — Split V into V_{even} (even degree) and V_{odd} (odd degree). Then $\Sigma_{V_{odd}} \deg(v) = 2|E| - \Sigma_{V_{even}} \deg(v)$. The RHS is a difference of two even numbers, hence even. A sum of odd numbers is even only when the number of terms is even. Hence $|V_{odd}|$ is even. 3
4. **Example** — degrees 3, 3, 2, 2, 2: sum = 12 = $2|E|$, so $|E| = 6$. Odd-degree vertices = 2 (even). ✓ 1